

series of commands and arguments that will be used in the next stage. The CPU, or the main processor, undertakes the task of creating the display list. In this stage, the CPU has to bring the required data closer to it, by moving it from the main memory to the cache memory.

The second stage is the conversion of the 3D scene to a 2D bitmap image. This is what video cards or GPUs are designed to do. Generally, this stage is referred to as rendering. Executing the display list creates the 2D bitmap image. The display list will contain tasks such as drawing lines and polygons, and giving textures and lighting. During this stage the GPU should be able to easily access the required data.

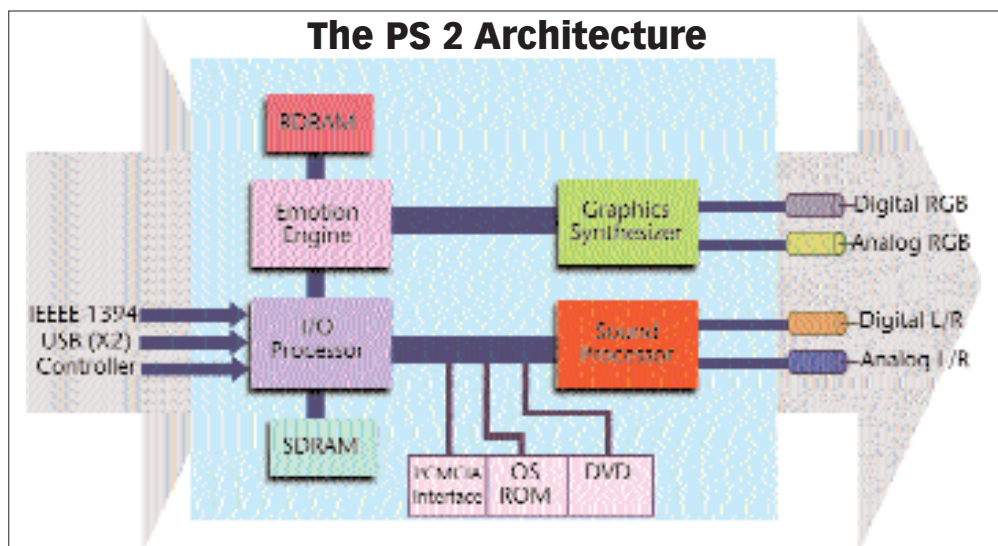
The final stage is the one that involves the least computation. Data is fetched from the frame buffer in a digital format, and converted into analog signals. These analog signals are sent to the monitor, which displays what you call 'stunning visuals'.

The consoles currently dominating the market are the Sony PS2, Microsoft Xbox and Nintendo GameCube. Here, we shall take a look at the first two. Read on to find out how they work.

PLAYSTATION 2: The almighty

The gaming console market is dominated by the PS2 to such an extent, that it has become the gamer's mantra. Sony recently unveiled their upgraded PS2, the PSX with additional features, aiming at making it a complete entertainment box. As with other gaming consoles, the designers of the PS2 had the luxury of building a system exclusively for gaming. The heart of the PS2 is the now famous 'Emotion Engine', which was jointly developed by Sony and Toshiba.

For all its boasts, the architecture of the PS2 is pretty simple. Little do people realise that it is a perfect example of KISS (Keep it Simple, Stupid). The main components of the PS2, in order of importance are, the Emotion Engine, Graphics Synthesizer, Sound Processor and I/O Processor. Let's look at the simplest one first. The I/O processor (IOP) handles all the requests from the outside world.



When the controller gives an input, the IOP sends it to the Emotion Engine, so that an update is made to the state of the game. The IOP is also responsible for running earlier PlayStation games. The sound processor is capable of producing CD quality audio to support the visuals. It supports formats such as AC3 and DTS—something that would be expected from a device of its class.

Coming to the serious stuff, the Emotion Engine has a 300 MHz MIPS III CPU as the core. Actually, the Emotion Engine comprises of many components, the important ones being a MIPS III CPU core, a FPU (Floating Point Unit), two vector units—VU0 and VU1, and a Graphics Interface Unit (GIF). These components are well connected. This is where PS2 scores over the rest of the pack. It has two advantages, namely bandwidth and strategically placed cache memory. The components of the Emotion Engine are connected to each other via a 128-bit bus. This is something unheard of in a PC. The reason why the bus width cannot be increased easily in a PC is that it has to maintain compatibility with existing applications. Increasing the speed will not affect legacy applications, but changing the bus width will.

The next difference is the cache memory. The sole purpose of the cache memory is to keep important data close at hand while processing. Hence, what matters more is the location of the cache memory, not just its size. The actual oper-

ations performed by the Emotion Engine depend upon the application. For example, in the case of a FPS game it has to execute the AI program for the enemy. In general, it deals with the behaviour of the game, such as AI and physics. But the ultimate output of the Emotion Engine is the display list.

The CPU teams up with the FPU and VU0, while the VU1 teams up with the Graphics Synthesizer (GS), through the GIF. The programmer decides the task assigned to these teams. They can be made to work in parallel and send the display list to the GIF together. Alternatively, the CPU group can be made to do some processing and send the result to the VU1, which will use the input and do some further processing and finally send the display list to the GIF. This parallel or serial option gives a lot of flexibility to the programmer.

The CPU has 16 KB of SPRAM (Scratch Pad RAM); this SPRAM can be used by the VU0 also. This piece of memory allows the CPU and VU0 to collaborate together. A 128-bit bus connects them to each other. Most of the time, the FPU and the VU0 act as co-processors to the MIPS CPU, but can be programmed to work separately. The VU0 performs operations such as floating-point divide, square root and inverse square root. The bulk of the processing that goes into 3D games involves doing a lot of these operations.

The VU1 acts more like a geometry processor for the GS. Since this involves more floating point calculations, it packs extra processing power and memory to do the same. Unlike the CPU team, where

info To put an end to finagling of the PS2 in India, Sony has announced that it would soon be launching the console in our country.